



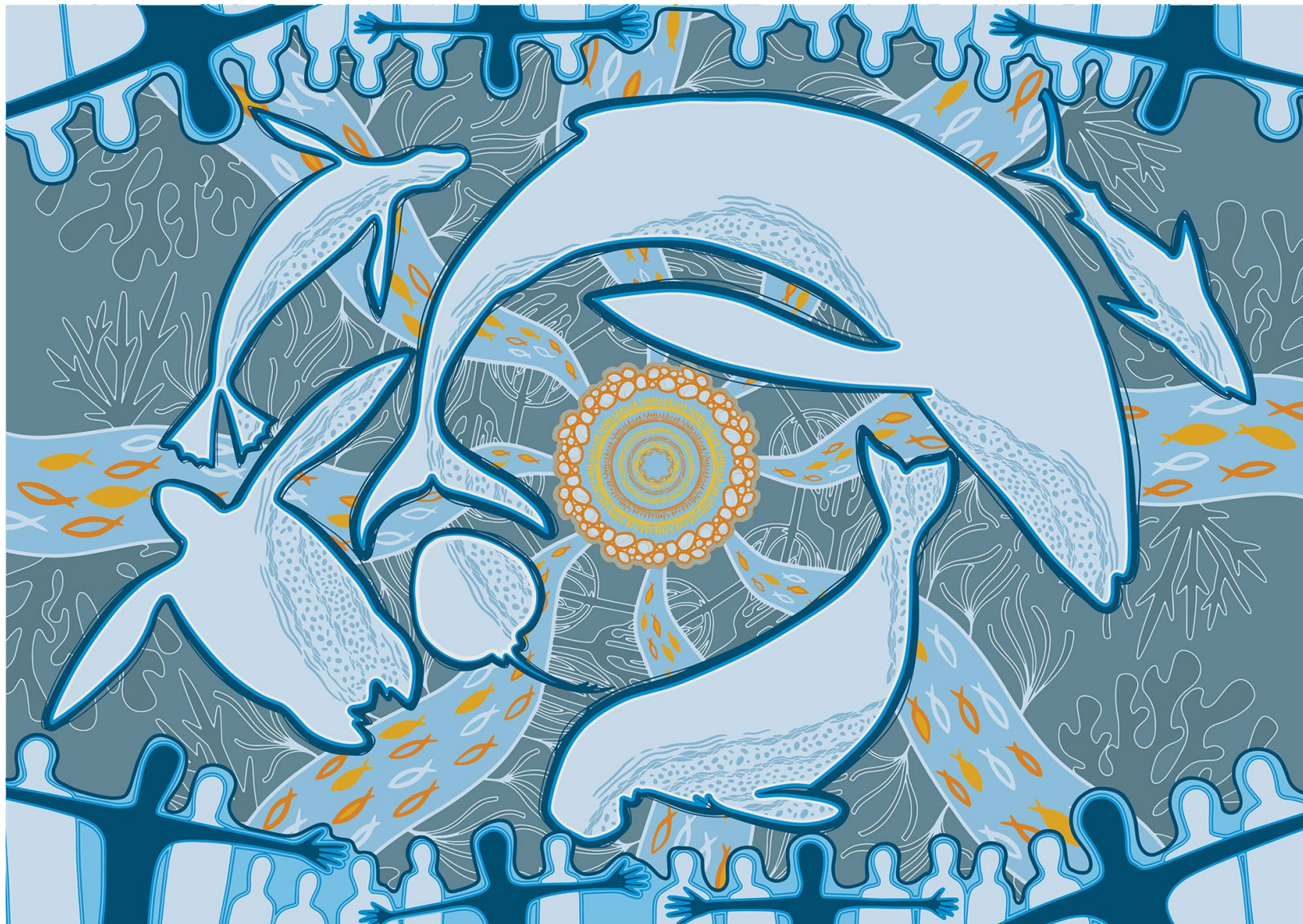
***Explore high-quality, open,
ocean data and tools to support
your marine science***

Acknowledgement of Country

IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work, and those of the land on which we meet today – the Wurundjeri Woi-Wurrung people and the Bununrong/Boon Wurrung people of the Kulin Nation.

We recognise their role as Australia's first marine scientists and carers of Sea Country.

We pay our respect to Aboriginal and Torres Strait Island peoples past and present.





Explore high-quality, open, ocean data and tools to support your marine science

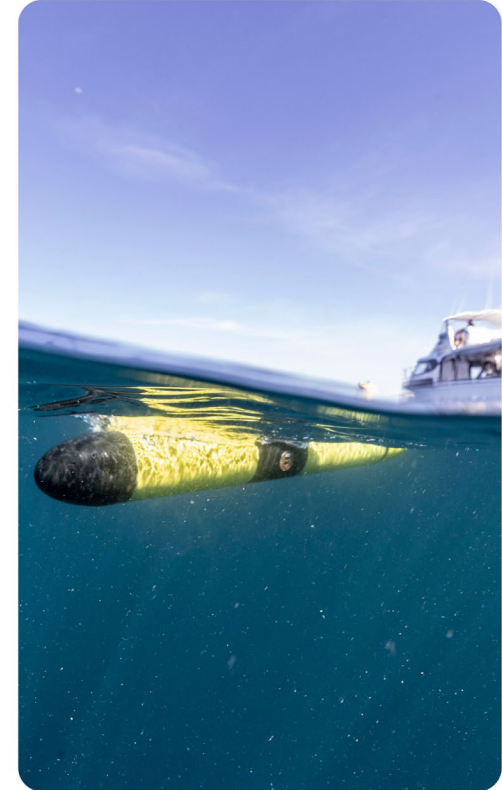
13th July 2025, Latrobe Ballroom, Sofitel Melbourne, 25 Collins Street



Time	Session	Presenters
8:30	Arrival & Registration	
9:00	Welcome & Acknowledgement of Country	
9:15	Australia's Integrated Marine Observing System (IMOS)	Fabrice Jaine
9:45	How reliable are IMOS Data?	Rebecca Zitoun
10:00	Where can I access IMOS data?	Mark Rehbein, Victoria Isaac, Benedicte Pasquer
10:30	Morning Tea	
11:00	IMOS OceanCurrent	Gabriela Semolini Pilo
11:30	IMOS Biological Ocean Observer	Claire Davies
12:30	Introducing IMOS cloud optimized datasets	Eduardo Klein
13:00	Lunch	
13:45	Guided tutorial: Using IMOS data in R	Denisse Fierro, Eduardo Klein
15:15	Afternoon Tea	
15:45	Guided tutorial: Using IMOS data in QGIS	Benjamin Stepin
16:30	Workshop Close	

Australia's Integrated Marine Observing System (IMOS):

**Supporting best practice ocean science,
decision-making and industry operations
through sustained, high-quality and open data**



Fabrice Jaïne – IMOS Principal Science Officer
fabrice.jaine@utas.edu.au



IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

Australia's Integrated Marine Observing System

- A **national, collaborative, research infrastructure** funded by the Australian Government in 2007 to provide **systematic, sustained and scientifically-robust observations of the marine environment** for research, management and industry.
- **IMOS** is guided by community-driven science and implementation planning, organised through jurisdictional Nodes that represent stakeholder needs and underpin what we need to observe and where.



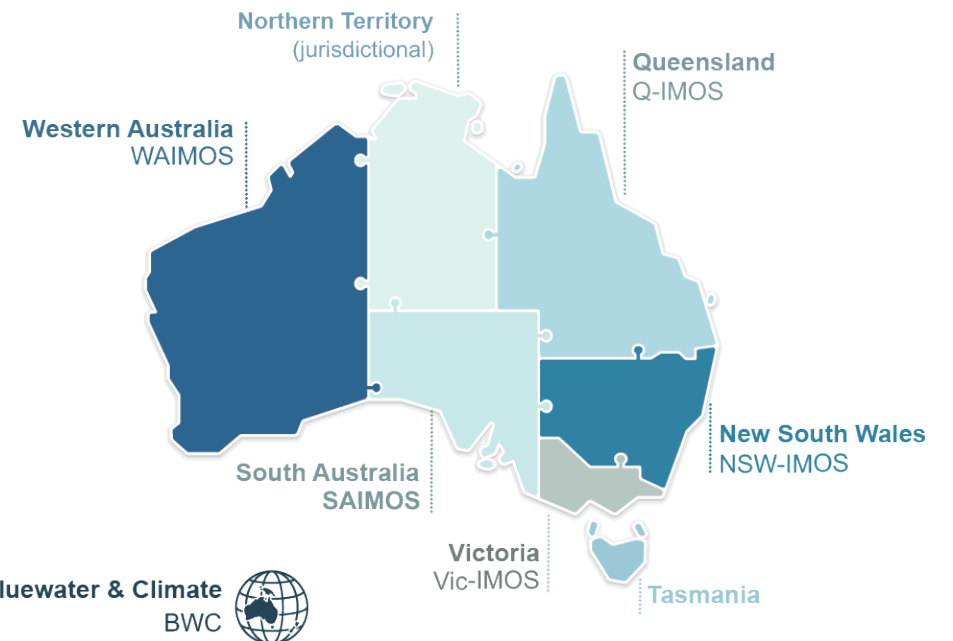
Australia's Integrated Marine Observing System (IMOS) is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent. www.imos.org.au

PRINCIPAL PARTICIPANTS



SIMS is a partnership involving four universities.

ASSOCIATE PARTICIPANTS



Australia's Integrated Marine Observing System

- A **national, collaborative, research infrastructure** funded by the Australian Government in 2007 to provide **systematic, sustained and scientifically-robust observations of the marine environment** for research, management and industry.
- **IMOS** is guided by community-driven science and implementation planning, organised through jurisdictional Nodes that represent stakeholder needs and underpin what we need to observe and where.
- IMOS is a Regional Alliance of the Global Ocean Observing System (GOOS) and is regarded as a leader in ocean observing, best practice protocols and data management.



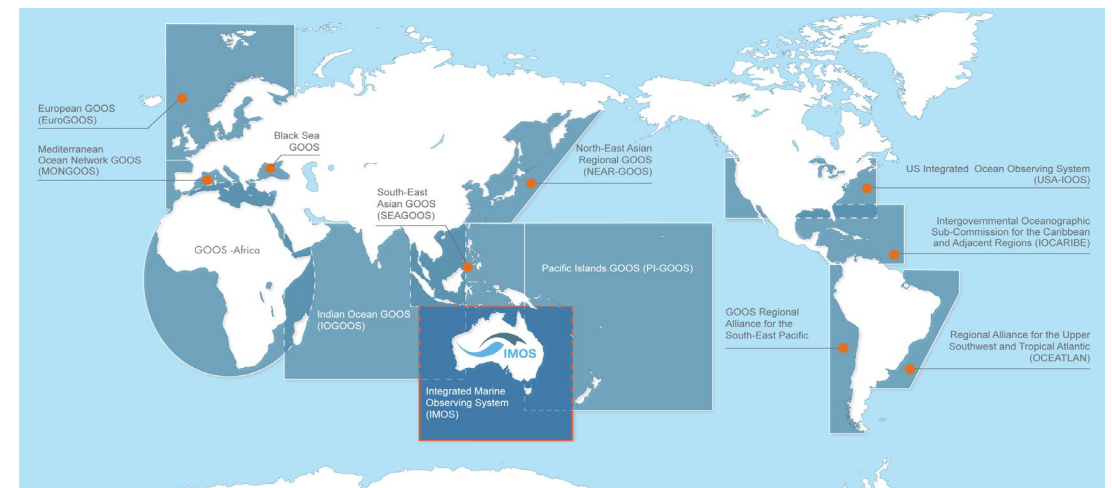
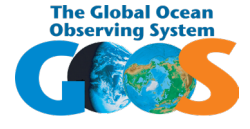
Australia's Integrated Marine Observing System (IMOS) is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent. www.imos.org.au

PRINCIPAL PARTICIPANTS



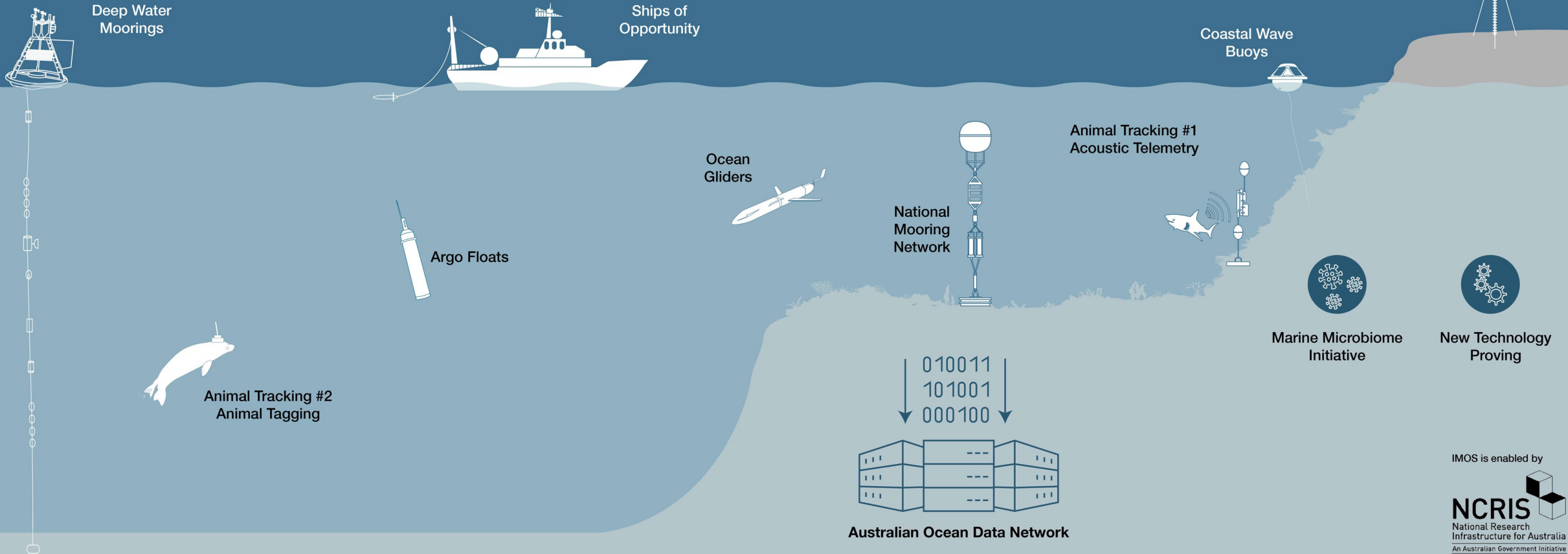
SIMS is a partnership involving four universities.

ASSOCIATE PARTICIPANTS



- Platform or Technology-based capability portfolio
- 14 Facilities, 52 sub-Facilities
- IMOS data are collected according to Best Practices in the field
- IMOS data are FAIR (Findable, Accessible, Interoperable, Reusable)

Our Facilities include



Ships of Opportunity

IMOS operates various sensors onboard a range of SOOPs to observe the ocean along their routes, from the poles to the tropics, and from coastal waters to the deep ocean.

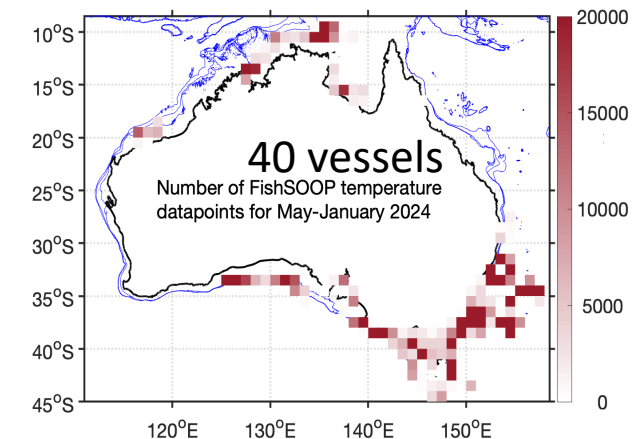
Research Vessels



Fishing Vessels

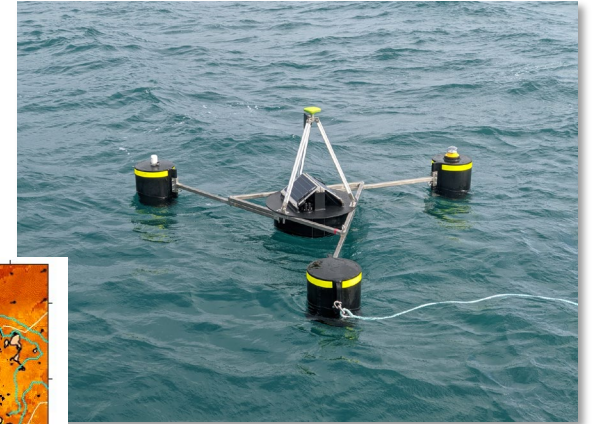
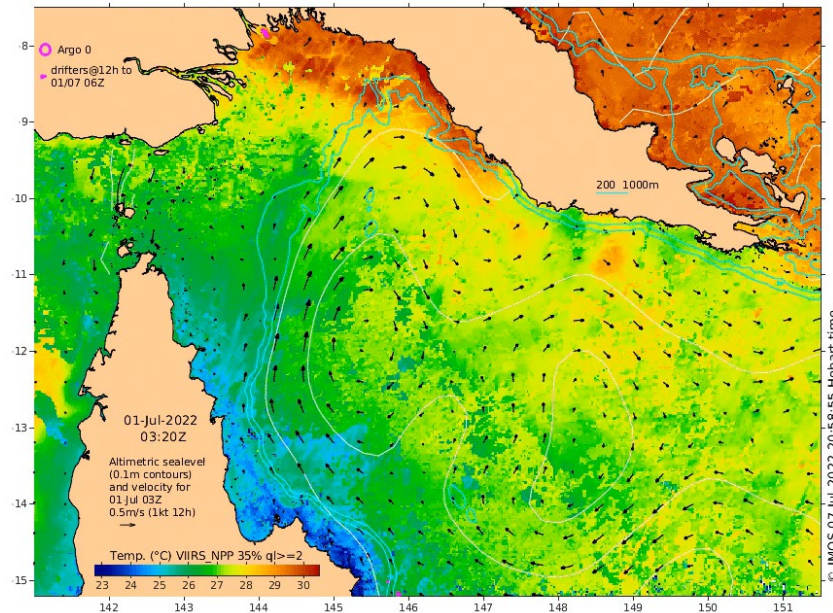


Industry Vessels



Satellite Remote Sensing

IMOS collaborates with various space agencies around the world to collect time series of *in situ* ocean temperature, colour and altimetry to validate and deliver calibrated satellite imagery.



Moored Instruments

IMOS also produces some of the longest ocean data time series in the Southern Hemisphere by deploying oceanographic moorings in coastal, shelf and offshore waters.

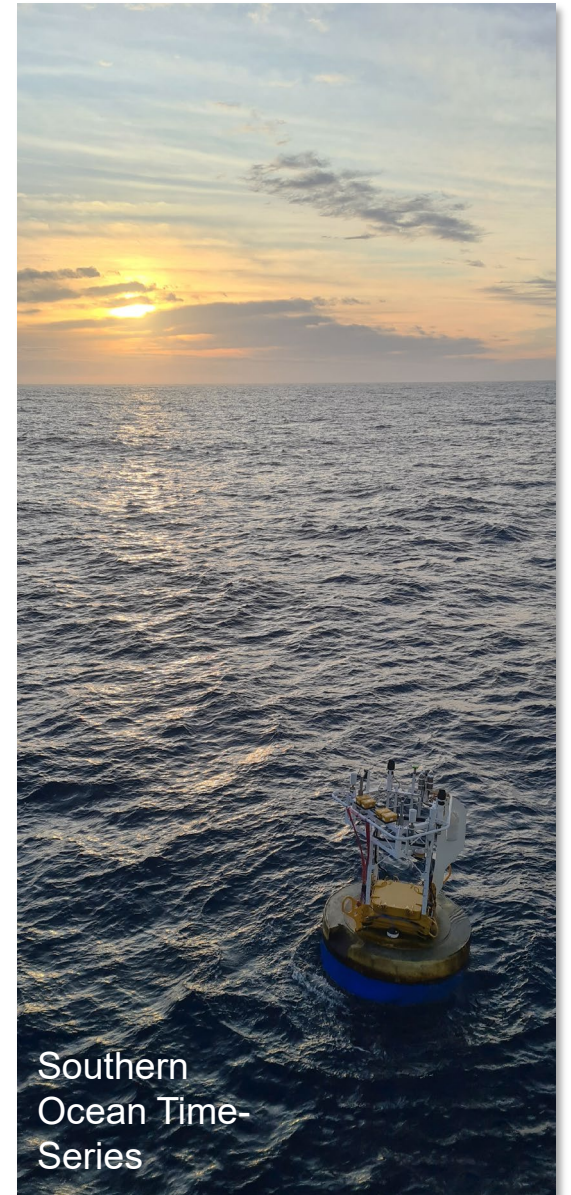
National References
Stations



Coastal Waves
Buoys

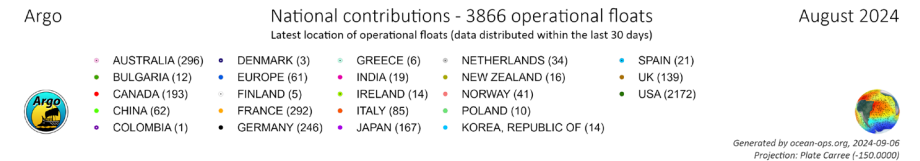
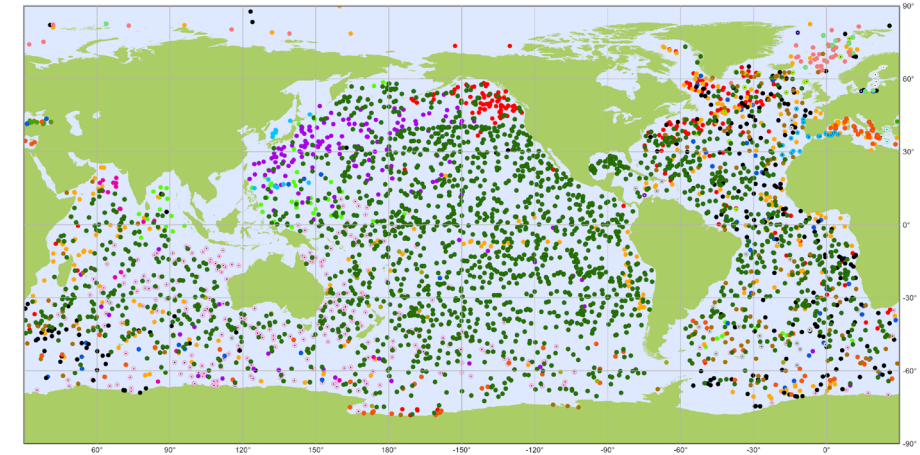


Southern
Ocean Time-
Series



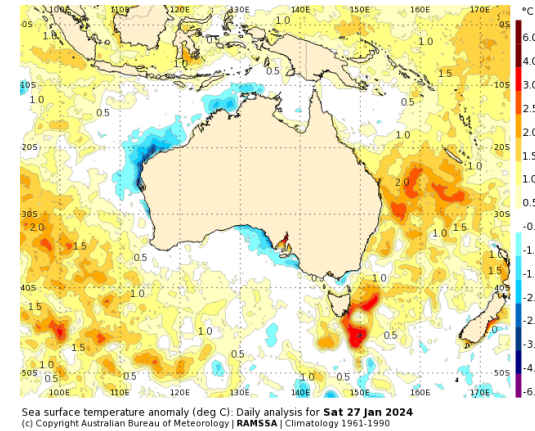
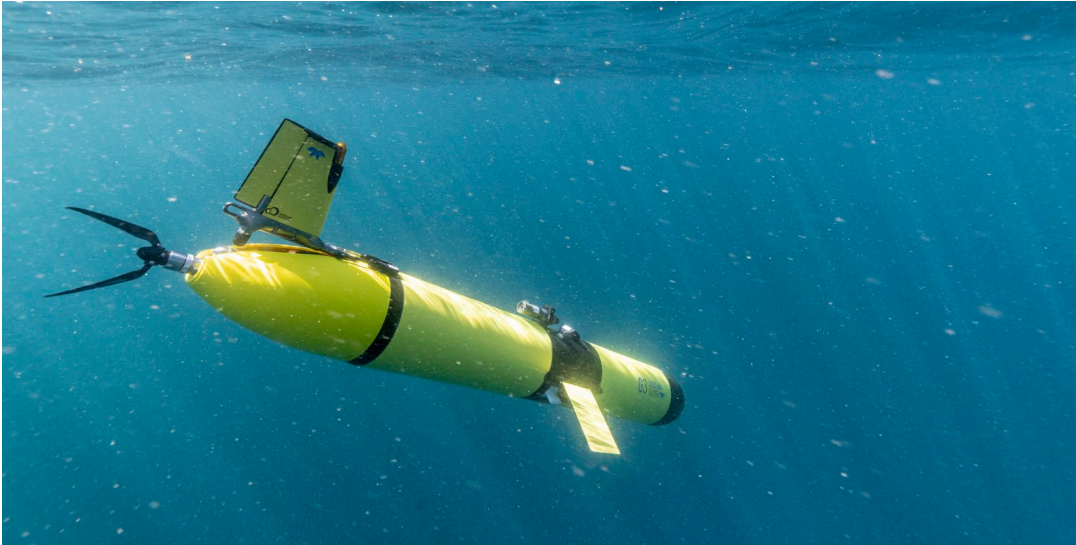
Argo Floats

IMOS is the second largest contributor to the global Argo program, which contributes real-time ocean observations to inform global weather forecasting and climate predictions.



Ocean Gliders

IMOS deploys ocean gliders in shelf waters around Australia to understand sub-surface characteristics of water masses, including during extreme weather events such as marine heatwaves, tropical cyclones, river run-off, etc.

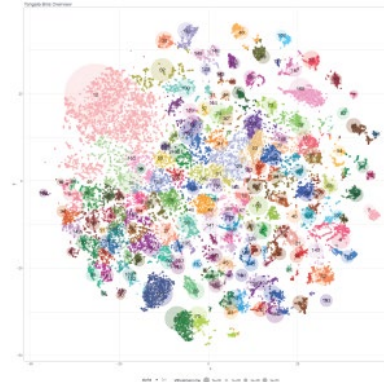


Water Sampling

IMOS conducts regular water sampling to understand patterns of ocean productivity, health and pollution.



Plankton



Nutrients, Microbes
& eDNA

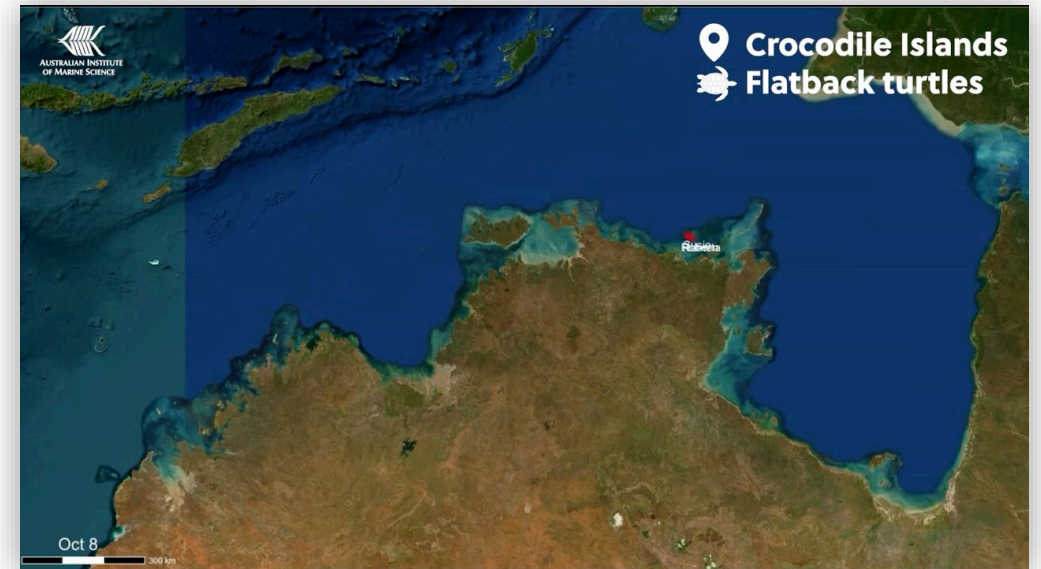
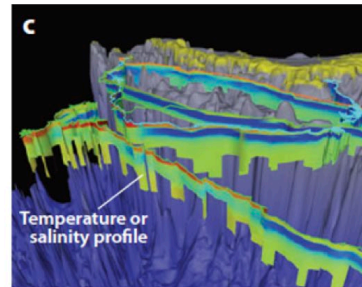
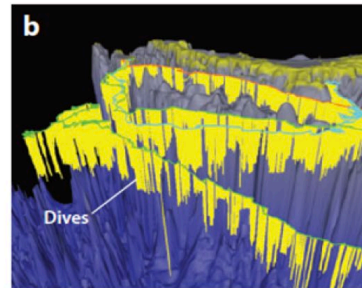
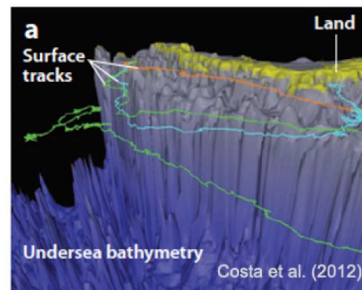


Microplastics



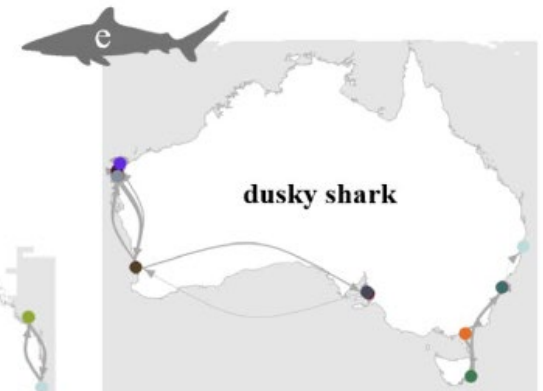
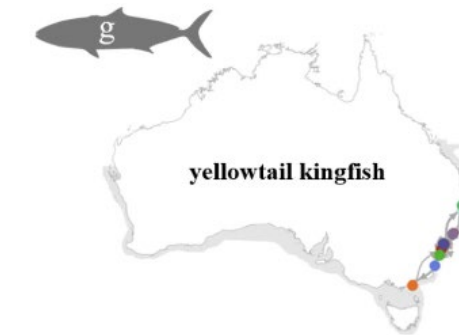
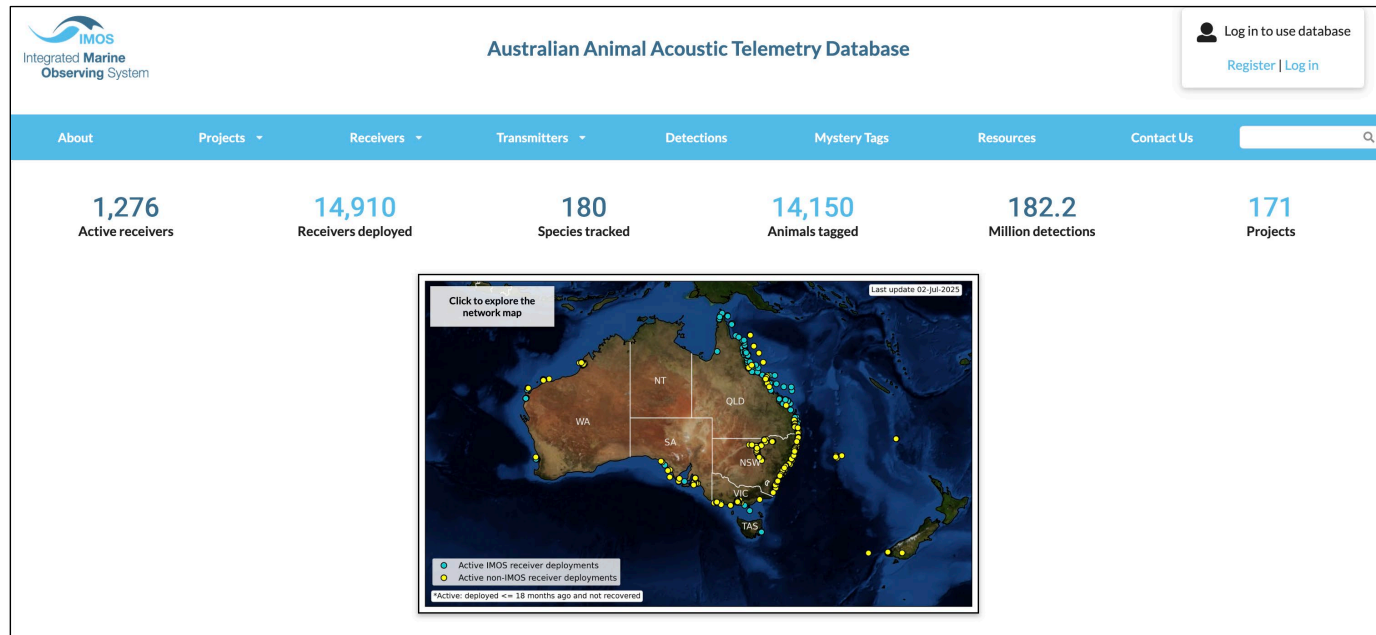
Animal-Borne Sensors

IMOS uses animals as oceanographers in places where conventional oceanographic instruments can't be deployed.

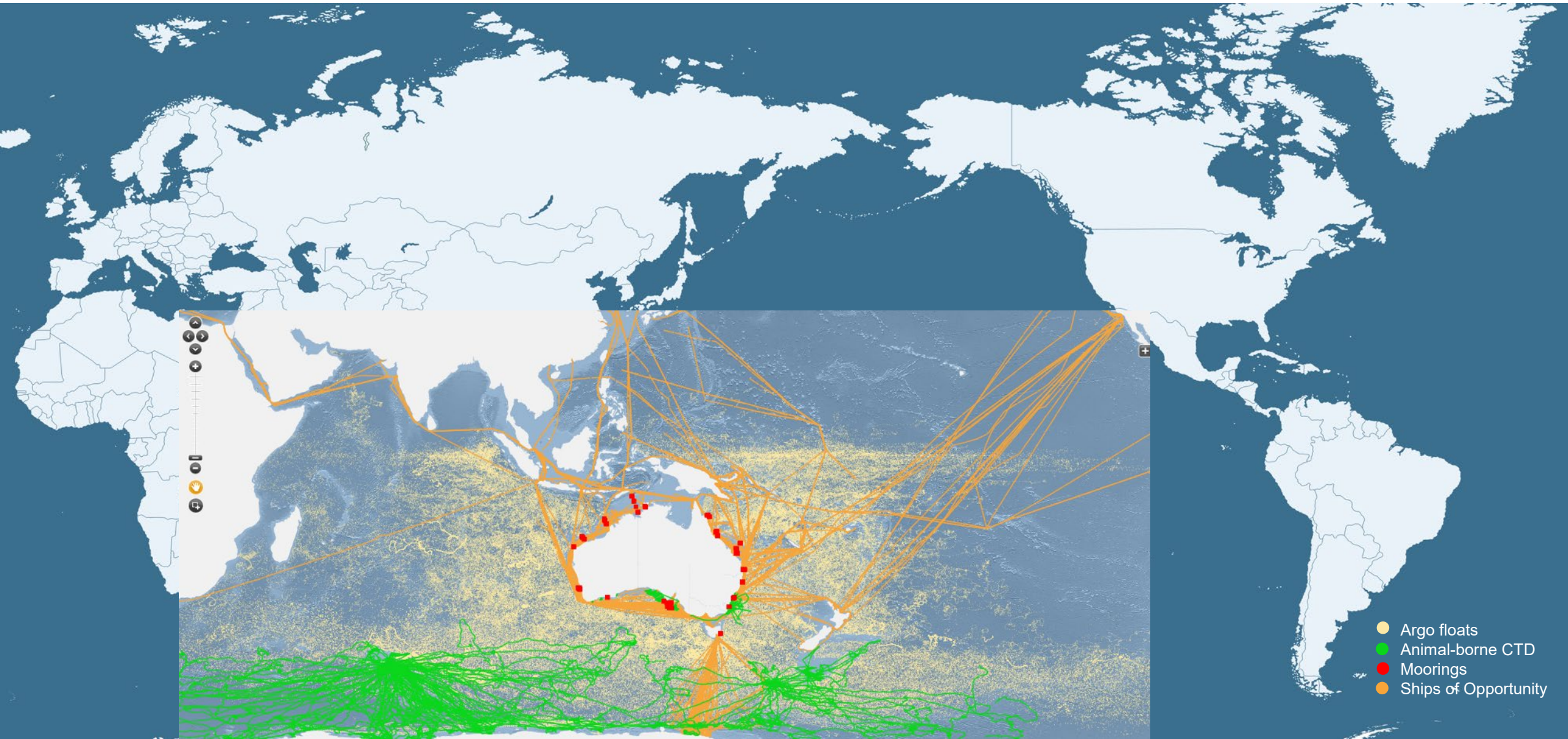


Animal Acoustic Tracking

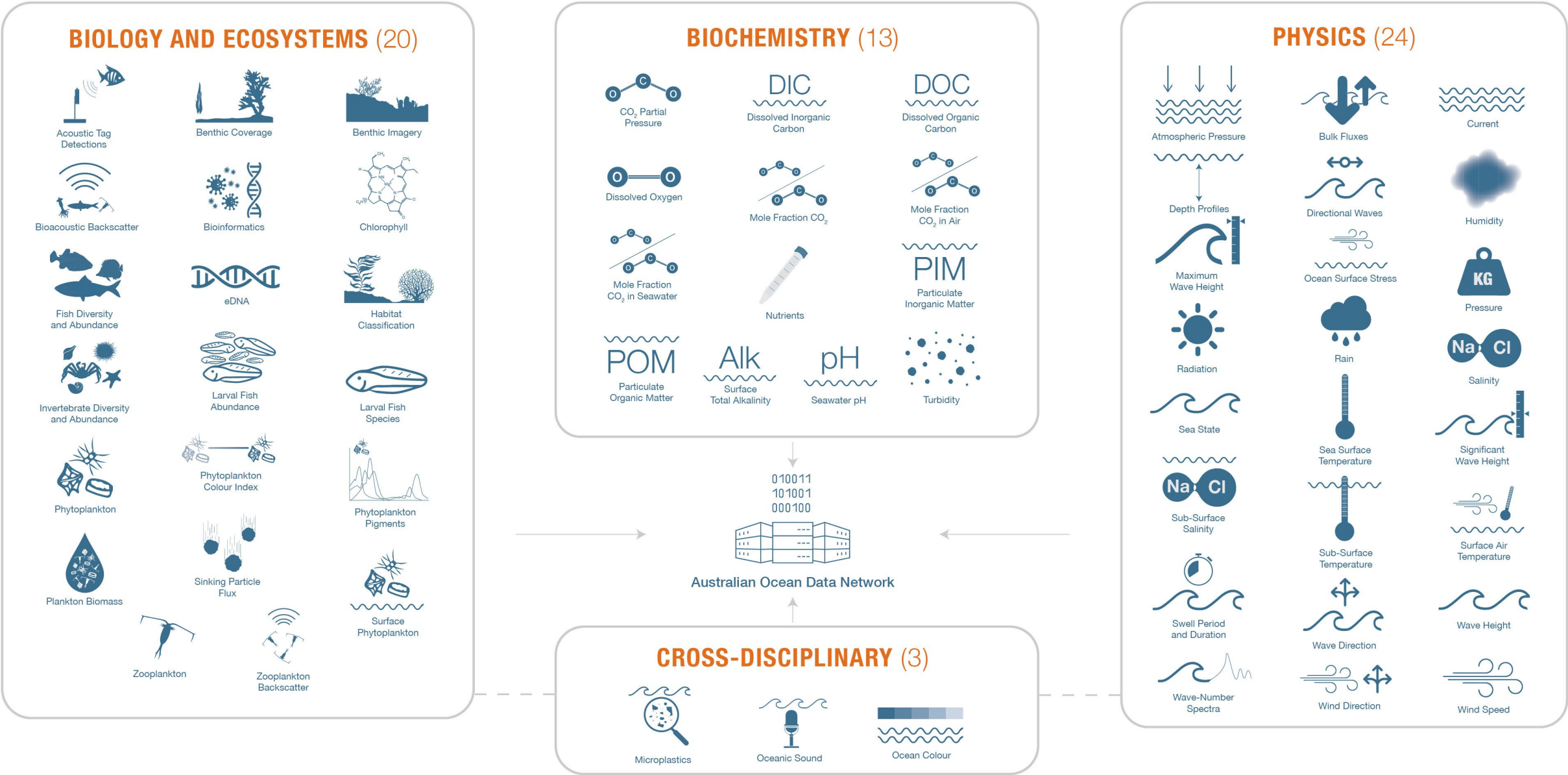
IMOS uses underwater hydrophones to detect broad-scale movements and connectivity of aquatic animals equipped with acoustic transmitters.



The IMOS operational footprint



IMOS delivers over 60 data streams contributing to 27 Essential Ocean Variables and 21 Essential Climate Variables

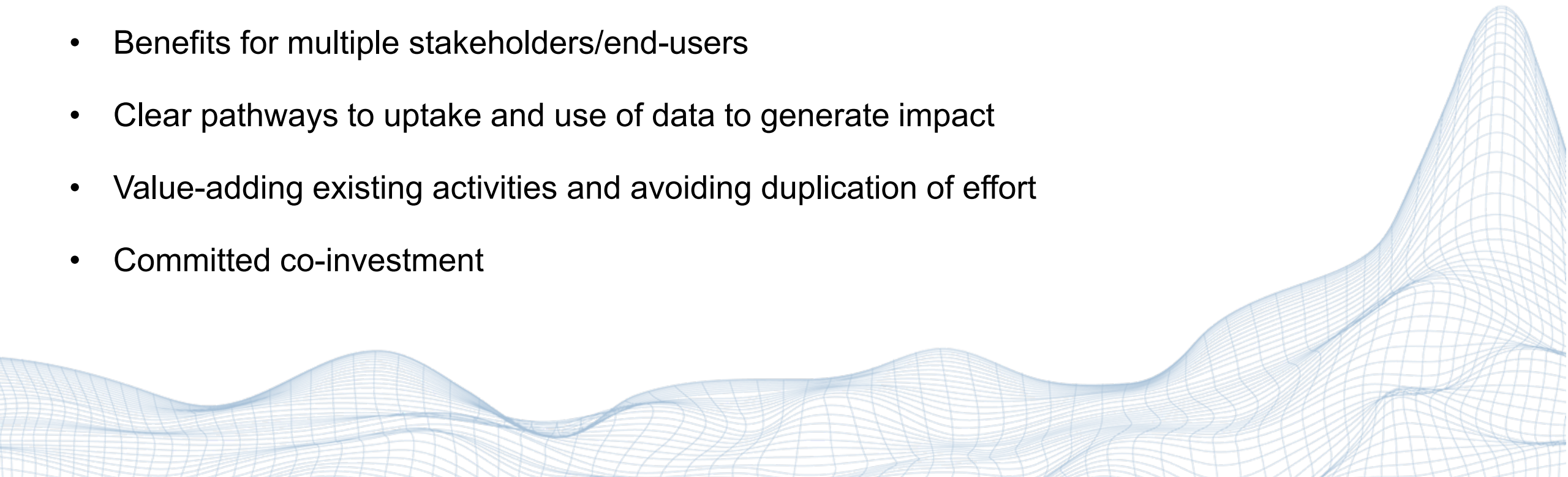


Since its inception in 2007,
IMOS has delivered over
**One Billion Ocean
Datasets**

How sustained are IMOS observations?

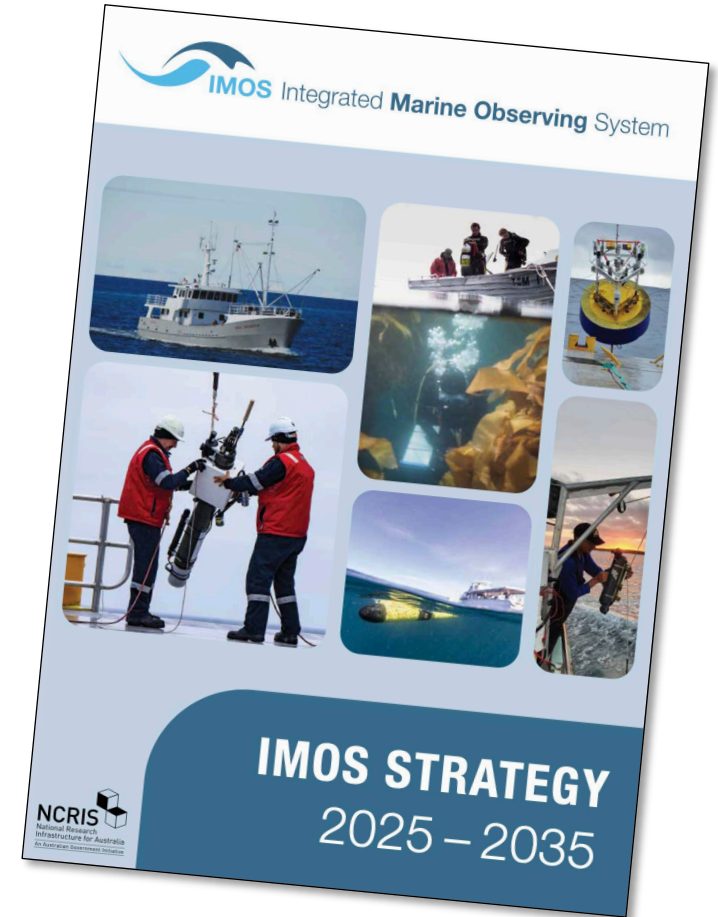
- IMOS has been funded by NCRIS since 2007 and has grown from strength to strength.
- **NCRIS funding currently secured to 2028:**
 - Continued baseline operations (11 Facilities, 47 sub-Facilities)
 - Step-change observing:
 - New (3) or enhanced observations
 - Data uplift
 - Coastal research infrastructure
- This new funding is enabling a step-change in IMOS's capacity to deliver real-time observations and data products to address stakeholder needs and inform science, operations and decision-making.

Criteria for prioritising IMOS investment

- Align with IMOS strategy and national priorities
 - Mature observing approaches, proven technology
 - Deliver long-term, sustained observations
 - Benefits for multiple stakeholders/end-users
 - Clear pathways to uptake and use of data to generate impact
 - Value-adding existing activities and avoiding duplication of effort
 - Committed co-investment
- 

Strategic Objectives

1. **Sustain and expand IMOS' world-leading ocean observing capability and data delivery** to support healthy oceans, coastlines and communities.
2. **Empower scientific discovery and unite communities** to address the impacts of ocean-driven climate change.
3. **Provide data** to strengthen Australia's blue economy and inform industry operations, national security, and government decision-making.
4. **Achieve greater impact** through partnerships with First Nations, industry, and others.
5. **Play a leadership and stewardship role** in regional and global ocean observing.
6. Through integration and collaboration, **produce data and data products to promote a better ocean future.**



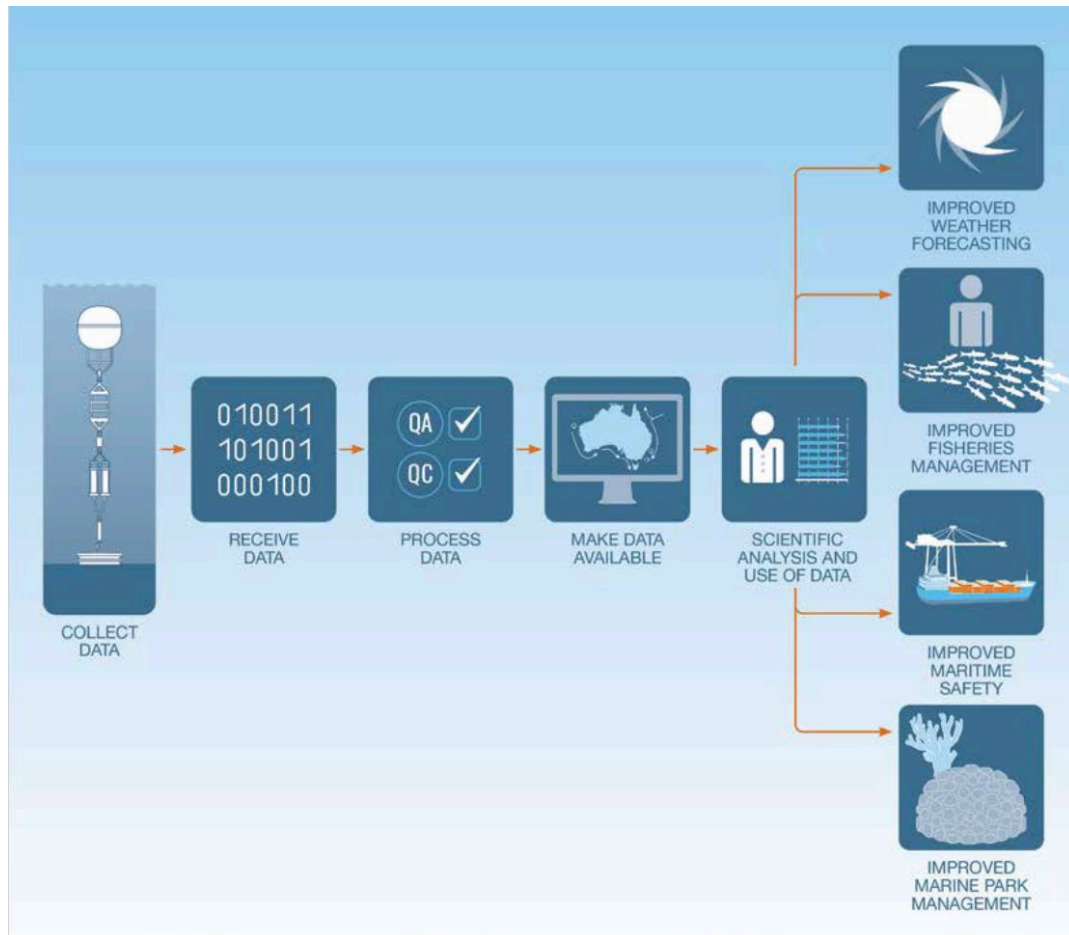
STATEMENT OF PURPOSE



Supporting solutions to environmental, climate, social and economic challenges at national and global scales through world-leading, sustained marine and Sea Country observing.

Tracing ocean observations to real-world outcomes

IMOS data are freely discoverable and accessible for use by the marine and climate science community, marine industries, policy-makers, First Nations, international agencies and other end-users to create societal benefit and impact for Australia.



THE NEED FOR OCEAN OBSERVING

IMOS delivers ocean observations with major benefits including:



Weather forecasting



Sea Country observing partnerships



Natural disaster and extreme weather preparedness



Offshore energy operations and maritime transport



Australian defence capability



Management of commercial fisheries



Improved policy responses



Use & Uptake

ALL of the **Top 20* Australian Universities** and
95% of the Top 20* World Universities
 have published with IMOS data
 *Based on QS World University Rankings 2025

4,493

Publications

514

Policy and Policy-related Documents

517

Postgraduate Projects

237

Products

754

Projects

3,299

Presentations

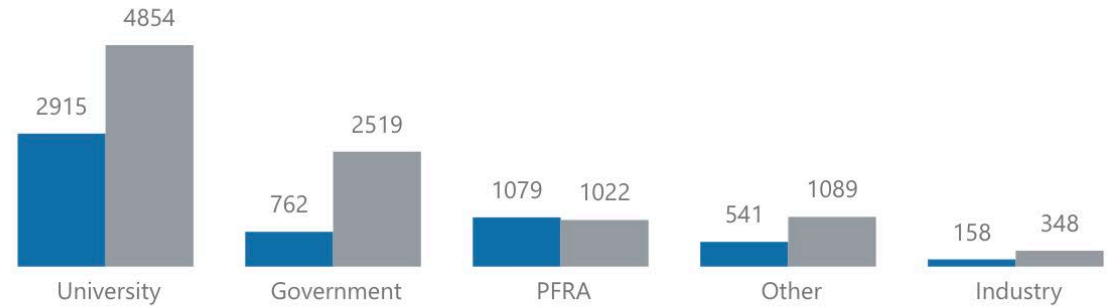
13,759 People across **2,551 Institutions**
 in **100 Countries** have used IMOS data

Use of IMOS data across the globe



Users by Type and Region

● Australian ● International



Publications contributing to **Societal Benefit Areas**



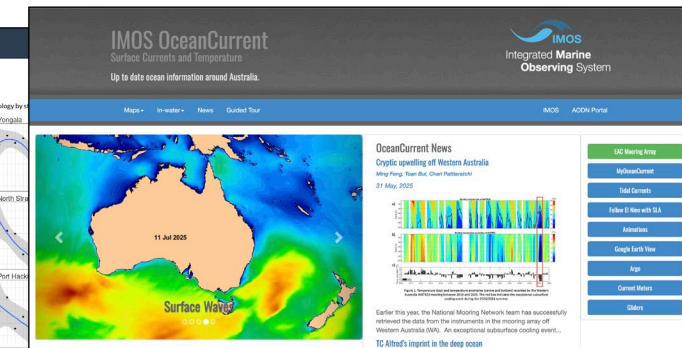
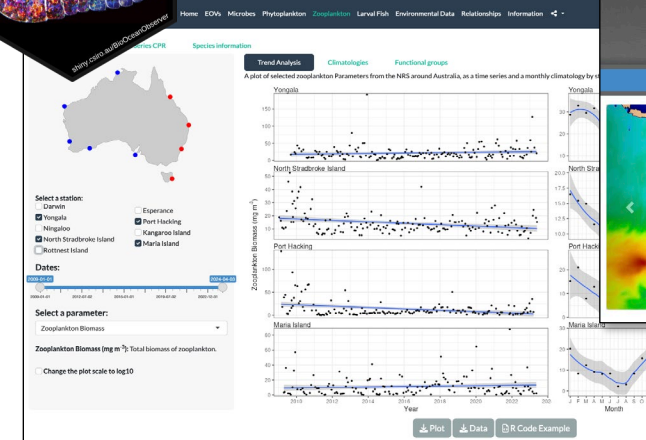
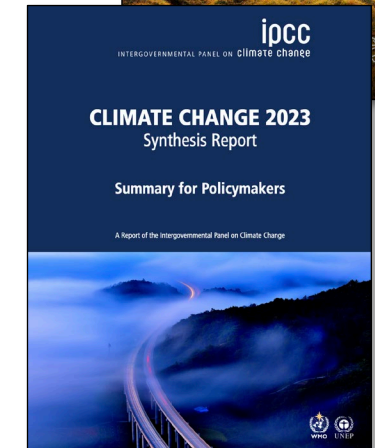
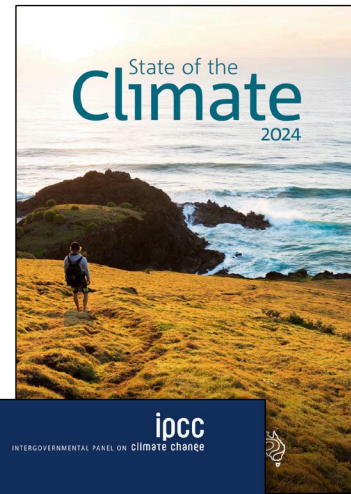
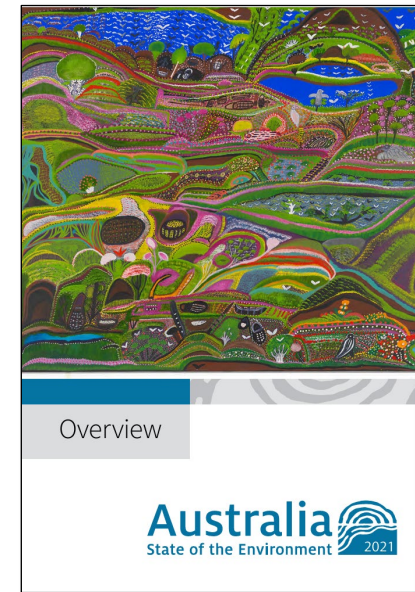
Publications contributing to

SUSTAINABLE DEVELOPMENT GOALS



IMOS Data Uses

- IMOS data have played an important role in Australia's State of the Environment and State of the Climate reporting, Intergovernmental Panel on Climate Change projections and report and numerous policy and management documents.
- IMOS observations contribute to a range of national ocean and climate models providing forecasting services for industry operations, defence capability and extreme weather preparedness.
- IMOS data products help deliver information to decision makers.



IMOS 20th Anniversary – Special Issue

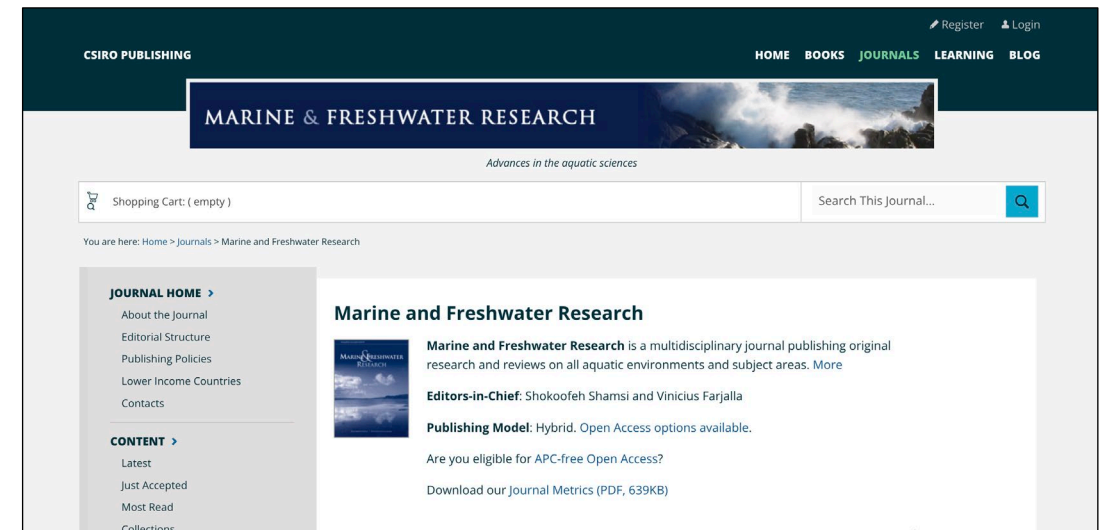
“Two decades of transformative ocean observations by Australia’s Integrated Marine Observing System (IMOS)”

Marine and Freshwater Research

Open-Access manuscripts showcasing the breadth of IMOS observations, uses and knowledge gained through 20 years of observing Australia’s marine estate.

Submissions: August – December 2025

Publication: 2026





Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS



SIMS is a partnership involving four universities

ASSOCIATE PARTICIPANTS



IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian, Western Australian and Queensland State Governments.

Get in touch:
fabrice.jaine@utas.edu.au

